

Deficiency of the Interleukin-1 Receptor Antagonist (DIRA): A Comprehensive Disease Characterization

MONDO:0013021 · OMIM #612852 · Orphanet ORPHA:361 · IL1RN (2q14.1)

Evidence sources are labeled: [Human clinical], [Model organism], [In vitro], [Review/aggregated]. All claims are cited to PMIDs; verbatim abstract quotes are given in quotation marks.

Summary (Answer to the Research Question)

DIRA is a rare, autosomal-recessive **autoinflammatory disease** caused by biallelic **loss-of-function mutations in *IL1RN***, the gene encoding the secreted interleukin-1 receptor antagonist (IL-1RA). Without IL-1RA to competitively block the type I IL-1 receptor (IL1R1), IL-1 α and IL-1 β signal unopposed, producing **NF- κ B-driven, neutrophil-mediated sterile inflammation** that classically presents in the **neonatal period** with **multifocal osteomyelitis, periostitis, and pustulosis** plus markedly elevated acute-phase reactants. It is life-threatening if untreated, but **IL-1 blockade with anakinra** (recombinant IL-1RA) produces rapid and durable remission in the large majority of patients — making DIRA a paradigm of mechanism-based targeted therapy. DIRA was first defined in 2009 (Aksentijevich et al., *NEJM*,

P 19494218

[Human clinical].

1. Disease Information

Overview. DIRA is a monogenic autoinflammatory disorder of skin and bone. Affected neonates/infants present with sterile multifocal osteomyelitis, periostitis, and pustular dermatitis with systemic inflammation, in the absence of infection, high-titer autoantibodies,

or autoreactive T cells [Human clinical,

P 19494218

Review

P 40060136

Key identifiers. - **MONDO:** MONDO:0013021 - **OMIM:** #612852 (disease); gene *IL1RN* 147679 - Orphanet: ORPHA:361 ("Deficiency of interleukin-1 receptor antagonist") - ICD-11: 4A60.x (autoinflammatory disorders) — *DIRA has no unique legacy ICD-10 code; typically coded under M08/M86 (osteomyelitis) or L98 historically* - MeSH: *DIRA is captured under "Hereditary Autoinflammatory Diseases" (MeSH D056660); MeSH descriptor used in the systematic review was "interleukin-1 receptor antagonist deficiency"*

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- HGNC: *IL1RN* = HGNC:6000; UniProt:* P18510

Synonyms / alternative names. Deficiency of interleukin-1 receptor antagonist; IL-1RA deficiency; IL-1 receptor antagonist deficiency; DIRA; osteomyelitis, sterile multifocal, with periostitis and pustulosis (OMPP).

Information source. Disease-level knowledge derives from **aggregated primary case reports and small case series** plus one PRISMA systematic review (18 studies, 2009–2024;

P 40060136

— not from large EHR/registry datasets, reflecting the disease's rarity (~30–60 reported genetically confirmed patients worldwide as of the mid-2020s).

2. Etiology

Primary cause — genetic. DIRA is caused by **autosomal recessive, biallelic loss-of-function mutations in *IL1RN*** (2q14.1, within the 2q13 IL-1 gene cluster). "DIRA is developed from the loss-of-function biallelic mutations of the *IL1RN* gene that encodes IL-1 receptor antagonist (IL-1RA), leading to the unchecked pro-inflammatory signaling and subsequent systemic inflammation"

[Review,

P 38398338

Genetic risk factors. - *Causal variants:* homozygous or compound-heterozygous null *IL1RN* alleles (nonsense, frameshift, large genomic deletions). Founder alleles cluster geographically (see §9). - *Modifier genes:* In the mouse model, disease penetrance/expressivity is strongly **background-dependent** and governed by QTLs (chromosome 1 major locus; candidate genes *Mr1*, *Pla2g4a*, *Fasl*, *Prg4*, *Ptgs2*) [Model organism,

P 21414240

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P 22942082

Human modifiers are not yet defined.

Environmental risk factors. None established as causal. **Consanguinity** and descent from a founder population increase the chance of being homozygous (see §9). Sex: reported **male predominance** [Review,

P 40060136

mechanism unknown (not X-linked; *IL1RN* is autosomal, so this is an observational skew).

Protective factors. Heterozygous carriers are asymptomatic — "heterozygous carriers were asymptomatic, with no cytokine abnormalities in vitro" [Human clinical/In vitro,

P 19494218

No environmental protective factors are described. The single functional protective "factor" is a wild-type *IL1RN* allele.

Gene–environment interactions. Not specifically characterized in humans. In *Il1rn*-null mice, the **intestinal microbiome** modulates disease: dysbiosis (↑ *Helicobacter*, ↓ *Ruminococcus/Prevotella*) promotes IL-17/TLR4-dependent arthritis, indicating a microbiome × genotype interaction [Model organism,

P 28645307

Whether infections trigger human DIRA flares is inferred but not proven.

3. Phenotypes

DIRA is dominated by **bone**, **skin**, and **systemic inflammatory** phenotypes with **neonatal onset** and **severe, chronic/episodic** course; frequencies below are qualitative/derived from the small literature

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Phenotype	Type	Onset	Severity	Frequency	HPO term
Multifocal sterile osteomyelitis	Clinical/ imaging sign	Neonatal	Severe	Very frequent (hallmark)	HP:0002754 (osteomyelitis)
Periostitis (periosteal reaction)	Imaging sign	Neonatal	Severe	Very frequent (hallmark)	HP:0002758- related / HP:0100774
Pustulosis / pustular rash	Physical manifestation	Neonatal– infancy	Moderate– severe	Very frequent	HP:0200039 (pustule); HP:0000988 (skin rash)
Fever / systemic inflammation	Symptom	Neonatal	Variable	Most common symptom	HP:0001945 (fever)
Osteolytic bone lesions	Imaging sign	Neonatal	Severe	Frequent	HP:0002797 (osteolysis)
Bone pain / pseudoparalysis	Symptom	Neonatal	Severe	Frequent	HP:0002653 (bone pain)
Widened/ballooned ribs, clavicles, long bones	Imaging sign	Neonatal	—	Characteristic	HP:0000892 (rib abnormality)
Nail changes (onychia, pustular paronychia)	Physical sign	Infancy	Mild– moderate	Frequent	HP:0001597 (nail dysplasia)
Arthralgia / arthritis / joint swelling	Sign	Neonatal– childhood	Variable	Frequent	HP:0002829 / HP:0001369
Respiratory distress / interstitial pneumonitis	Sign	Neonatal	Severe	Occasional– frequent	HP:0002098
Venous / arterial thrombosis, vasculitis	Complication	Variable	Severe	Occasional	HP:0002625 (deep vein thrombosis)

Phenotype	Type	Onset	Severity	Frequency	HPO term
Hepatosplenomegaly	Sign	Variable	—	Occasional	HP:0001433
Elevated CRP/ESR, neutrophilic leukocytosis, thrombocytosis, anemia	Laboratory abnormality	Neonatal	—	Near- universal	HP:0011227; HP:0001974; HP:0001894; HP:0001903

Supporting quotes: "neonatal onset of sterile multifocal osteomyelitis, periostitis, and pustulosis" [PMID 19494218]; "most common symptoms were fever, followed by osteoarticular manifestations (arthralgia, muscle contracture, fracture, osteolytic lesions, and osteomyelitis), nail changes, pneumonia, venous thrombosis" [PMID 40060136]; skin biopsy shows "neutrophil infiltration in the epidermis and subepidermal pustular dermatosis" [PMID 26100510].

Onset variability. Although classically neonatal, **late-onset** DIRA occurs (first symptoms at age 1–12 years) with milder or atypical presentations [Human clinical,

P 26100510

P 31467740

Quality-of-life impact. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease. Qualitatively, untreated DIRA causes severe pain, feeding difficulty, disability from bone deformity/fracture, and high mortality; effective IL-1 blockade dramatically improves QoL and prevents irreversible complications [Review,

P 32882069

4. Genetic / Molecular Information

Causal gene. *IL1RN* (HGNC:6000; NCBI Gene 3557; Ensembl ENSG00000136689; OMIM 147679), chromosome 2q14.1, encoding IL-1 receptor antagonist (IL-1RA; UniProt P18510)*, a secreted competitive antagonist of IL1R1.

Pathogenic variants (all loss-of-function; germline; ACMG "Pathogenic"): - **p.E77X** (nonsense) — Newfoundland founder allele [PMID 19494218, 21792839] - **2-bp deletion** (frameshift) — Netherlands founder allele [PMID 19494218] - **Lebanese frameshift** (consanguineous family) [PMID 19494218] - **~175-kb genomic deletion at 2q13** removing *IL1RN* + five other IL-1-family genes — Puerto Rican founder allele [PMID 22431714, 22431772] - **c.140delC / p.T47TfsX4** (novel frameshift) in compound het with E77X [PMID 21792839] - **p.Asp72_Ile76del** (in-frame 15-bp deletion) — recurrent in Brazilian patients; **p.Q45*** (rs1019766125) nonsense [PMID 32819369] - **22,216-bp deletion** spanning the first four *IL1RN* exons — Indian patient, likely founder [PMID 28503715] - **p.R26X** (nonsense) — late-onset Turkish patient [PMID 26100510]

Variant classes: nonsense, frameshift, in-frame deletion, and **large structural/genomic deletions**. Because large deletions are common, **copy-number-aware testing (SNP array/CMA or MLPA)** is required in addition to sequencing.

Allele frequency. Individually ultra-rare in gnomAD; founder alleles are enriched in specific populations (e.g., high carrier frequency of the 175-kb deletion in Puerto Ricans — "A high carrier frequency of the 175-kb DIRA-associated genomic deletion in the Puerto Rican population..." [PMID 22431714]).

Somatic vs germline: exclusively **germline**, biallelic, recessive.

Functional consequence: Loss of function. "The *IL1RN* mutations resulted in a truncated protein that is not secreted, thereby rendering cells hyperresponsive to interleukin-1beta stimulation" [In vitro/Human,

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P 19494218

One most-upstream premature stop codon variant is **hypomorphic** due to translation reinitiation, softening the phenotype [In vitro,

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P 32185578

No dominant-negative or gain-of-function mechanism.

Modifier genes: human modifiers undefined; strong QTL modifiers in mice (\$2, \$15).

Epigenetic information: No DIRA-specific DNA-methylation/histone data available (not applicable/unknown).

Chromosomal abnormalities: the disease-relevant "chromosomal" lesion is the **contiguous-gene 2q13 microdeletion** (structural variant) that includes *IL1RN* [PMID 22431714].

GO/molecular annotations: *IL1RN* — **GO:0005152** (IL-1 receptor antagonist activity), **GO:0005615** (extracellular space), **GO:0002526** (acute inflammatory response, negative regulation).

5. Environmental Information

- **Environmental / toxic / occupational factors:** none causal (monogenic disease).
- **Lifestyle factors:** not applicable (neonatal onset).
- **Infectious agents:** DIRA inflammation is **sterile** — lesions are culture-negative with no organisms on histology [Review,

P 21788901

P 39757386

Infection is a key differential to exclude, not a cause. In the mouse model, the **commensal microbiome** modulates disease severity [Model organism,

P 28645307

suggesting microbial context may influence expressivity but is not the trigger.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic null *IL1RN* mutation** (nonsense/frameshift/deletion) **leads to** absent or truncated IL-1RA that **is not secreted** [In vitro,

P 19494218

2. Loss of secreted IL-1RA **results in** loss of competitive antagonism at the **type I IL-1 receptor (IL1R1)** on target cells.
3. Unopposed **IL-1 α and IL-1 β bind IL1R1 + accessory protein IL1RAP, leading to** recruitment of MyD88 and IRAK4/IRAK1–TRAF6 (canonical IL-1 signaling; GO:0070498).

4. This **results in** sustained activation of **NF-κB** and MAPK — demonstrated as "significantly enhanced and prolonged" nuclear translocation of **NF-κB p65** in IL-1RA-deficient tissue [Model organism/In vitro,

P 16622029

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5. NF-κB activation **drives transcription of** IL-6, IL-8/CXCL8, CXCL1(KC)/CXCL2(MIP-2), TNF-α, and G-CSF [Model organism,

P 16622029

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In vitro comparison

P 33314777

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6. Chemokine/cytokine output **leads to** massive **neutrophil recruitment** — "IL-1β is the pivotal cytokine... responsible for the exaggerated production of cytokines and chemokines that induce the recruitment of neutrophils" [Review,

P 29742056

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7. Branch A — **Skin:** neutrophils accumulate subcorneally **producing** pustulosis/psoriasiform dermatitis with "neutrophil-rich microabscesses... beneath the stratum corneum" [Model organism,

P 15086551

] → clinical **pustular rash**.
8. Branch B — **Bone:** IL-1 activates osteoclastogenic mediators and inflammatory macrophages (elevated caspase-1/IL-1β secretion shown in DIRA monocytes/macrophages) **leading to** osteoclast/osteoblast dysregulation, osteolysis, and periosteal new bone → **multifocal osteomyelitis + periostitis** [In vitro,

P 33314777

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9. Branch C — **Systemic/vascular:** circulating IL-1/IL-6 **produce** fever, acute-phase response (↑ CRP/ESR), and predispose to **thrombosis/vasculitis** (aortitis in the mouse model) [Model organism,

P 39600116

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Human

P 40060136

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10. Interrupting step 2–3 pharmacologically (anakinra/rilonacept/canakinumab) **restores** antagonism/ligand neutralization and **results in** rapid clinical remission [Human clinical,

P 19494218

P 38398338

— closing the causal loop and confirming IL-1 as the driver.

Upstream vs downstream. Upstream/primary: *IL1RN* LOF → unopposed IL1R1 signaling → NF-κB. Downstream/effector: chemokine amplification, neutrophil influx, osteoclast activation, tissue injury.

Cellular processes: innate immune activation, chronic sterile inflammation (GO:0006954), neutrophil chemotaxis (GO:0030593), NF-κB signaling (GO:0007249), osteoclast differentiation (GO:0030316). **Cell types (CL):** neutrophils **CL:0000775**, monocytes/macrophages **CL:0000235**, keratinocytes **CL:0000312**, osteoclasts **CL:0000092**, osteoblasts **CL:0000062**, fibroblast/synoviocytes. **Protein dysfunction:** loss of function of secreted IL-1RA (misprocessed/non-secreted truncation) [PMID 19494218]. **Metabolic changes:** none disease-specific. **Chemical entities (CHEBI):** IL-1β and IL-1α (proteins); therapeutic small-molecule/biologic antagonists; reactive mediators nitric oxide (CHEBI:16480), prostaglandins via PTGS2/COX-2. **Molecular profiling:** no large-scale omics datasets specific to DIRA patients (GEO/PRIDE/MetaboLights: none dedicated). Mechanistic transcriptomics comes from *Il1rn*-KO mice and comparative monocyte studies

P 33314777

P 39600116

7. Anatomical Structures Affected

- **Primary organs: Bone / skeletal system** (UBERON:0002481 bone tissue; ribs UBERON:0002228; clavicle UBERON:0001105; long bones/metaphyses; vertebrae UBERON:0002412) and **skin** (UBERON:0002097; epidermis UBERON:0001003; nail UBERON:0001705).
- **Secondary / systemic involvement:** joints (UBERON:0000982), **lungs/respiratory system** (interstitial pneumonitis; UBERON:0002048), **vasculature** (arteries/veins; thrombosis, aortitis UBERON:0000947), liver/spleen (hepatosplenomegaly), CNS rarely.

- **Body systems:** musculoskeletal, integumentary, immune/hematologic, respiratory, cardiovascular.
- **Tissue types:** connective/osseous tissue, epithelial (epidermis, nail matrix), synovium, vascular endothelium.
- **Cell populations:** neutrophils (CL:0000775), macrophages (CL:0000235), keratinocytes (CL:0000312), osteoclasts (CL:0000092).
- **Subcellular (GO CC):** extracellular space (GO:0005615, site of missing IL-1RA); plasma-membrane receptor complex IL1R1/IL1RAP (GO:0045323); nucleus (NF- κ B translocation, GO:0005634).
- **Localization / lateralization:** **multifocal and bilateral/asymmetric** bone lesions; diffuse/generalized skin pustulosis. Characteristic sites: ribs, clavicles, long-bone metaphyses, vertebrae, periarticular bone [Human clinical,

P 28503715

"sterile multifocal osteomyelitis including ribs and clavicles"].

8. Temporal Development

- **Onset:** typically **congenital/neonatal** (first days–weeks of life), acute-to-subacute; **variable**, with recognized **late-onset** forms (1–12 years) [PMID 19494218, 26100510, 31467740].
- **Progression:** **chronic, progressive or episodic/relapsing** systemic inflammation with flares; untreated disease escalates to organ failure. Progression rate can be **rapid** in neonates (respiratory insufficiency, sepsis-like deterioration) [PMID 37575641, 39757386].
- **Duration:** **chronic, lifelong** — requires continuous therapy; no spontaneous cure.
- **Remission pattern:** **treatment-induced** (IL-1 blockade); relapse on drug withdrawal. Spontaneous remission is not characteristic.
- **Critical period / window of opportunity:** **early initiation of anakinra** (ideally before irreversible bone/organ damage) is the decisive intervention window — "It is imperative to recognize this disease early to achieve adequate response and remission" [PMID 37575641].

9. Inheritance and Population

- **Inheritance: Autosomal recessive** [PMID 19494218].
- **Penetrance:** essentially **complete** in biallelic-null genotypes (hypomorphic reinitiation alleles can attenuate;
([P 32185578](#))).
- **Expressivity: variable** (neonatal severe → late-onset milder), influenced by residual protein and presumed modifiers.
- **Carriers:** asymptomatic, no in-vitro cytokine abnormality [PMID 19494218].
- **Founder effects (variant geography):** Newfoundland (E77X), Netherlands (2-bp del), Puerto Rico (175-kb 2q13 deletion; high local carrier frequency), Lebanon (frameshift), Brazil (p.Asp72_Ile76del), India (22.5-kb deletion) [PMID 19494218, 22431714, 32819369, 28503715].
- **Consanguinity:** contributory (e.g., Lebanese consanguineous family) [PMID 19494218].
- **Anticipation / germline mosaicism:** not described / not applicable.
- **Epidemiology: ultra-rare;** ~30–60 genetically confirmed cases reported worldwide; precise prevalence/incidence **not established** (Orphanet lists prevalence as unknown/ <1/1,000,000). Systematic review: 18 eligible studies over 15 years
([P 40060136](#))).
- **Sex ratio:** reported **male-predominant** (~observational;
([P 40060136](#)));
disease gene is autosomal.
- **Age distribution:** overwhelmingly infants; clinical recognition often around ~4 years in the aggregate literature
([P 40060136](#))),
reflecting diagnostic delay.

10. Diagnostics

- **Laboratory (LOINC):** markedly elevated **CRP** (LOINC 1988-5) and **ESR** (LOINC 4537-7); neutrophilic leukocytosis, thrombocytosis, anemia of inflammation [PMID 19494218, 26100510]. HPO: HP:0011227, HP:0001974, HP:0001894, HP:0001903.
- **Imaging (RadLex/Radiopaedia):** radiographs show **osteolytic and/or sclerotic lesions, periosteal new bone, rib/clavicle/long-bone widening, vertebral involvement, heterotopic ossification**; **MRI / whole-body MRI is the sensitivity gold standard** and detects clinically silent lesions — "MRI is more sensitive for detecting CNO and is considered the gold standard for monitoring the disease" [PMID 30031498]; "Whole body MRI is helpful in detecting asymptomatic lesions" [PMID 21788901].
- **Histopathology:** skin — subepidermal/subcorneal **neutrophilic pustular dermatosis** [PMID 26100510]; bone — **sterile osteomyelitis** with "predominant neutrophil infiltration in the absence of autoantibodies and autoreactive T cells" [PMID 21788901]. SNOMED: sterile/chronic nonbacterial osteomyelitis.
- **Genetic testing (definitive):** **single-gene *IL1RN* sequencing** or **NGS autoinflammatory panels / WES, plus CMA/SNP-array or MLPA** to capture large deletions (e.g., 175-kb, 22.5-kb) [PMID 22431714, 28503715]. WGS captures both SNVs and structural variants. GTR: *IL1RN*-testing labs available.
- **Functional/biomarker confirmation:** demonstration of absent secreted IL-1RA and IL-1 β hyperresponsiveness of patient cells [In vitro,

P 19494218

- **Clinical criteria:** no formal consensus criteria; **diagnosis = compatible skin/bone/systemic phenotype + biallelic *IL1RN* LOF variants** ($\pm$ dramatic anakinra response, sometimes used empirically before genetics;

P 37575641

- **Differential diagnosis:** infectious osteomyelitis; **CRMO/CNO**; **Majeed syndrome (*LPIN2*)**; **NOMID/CAPS (*NLRP3*)**; **DITRA (*IL36RN*)**/generalized pustular psoriasis; **PAPA (*PSTPIP1*)**; cherubism; neonatal sepsis; Langerhans cell histiocytosis [PMID 21788901, 39757386, 33314777]. Discriminator: sterile, antibiotic-unresponsive, **anakinra-responsive** disease with biallelic *IL1RN* variants.

- **Screening:** targeted **cascade/carrier testing** in founder populations and relatives; consider *IL1RN* testing (incl. CNV) in any neonate with unexplained pustulosis in high-carrier regions [PMID 22431714]. Not on standard newborn-screening panels.

11. Outcome / Prognosis

- **Untreated: life-threatening;** high morbidity/mortality with respiratory insufficiency, thrombosis, sepsis-like deterioration, multi-organ failure [PMID 38398338, 37575641, 39757386].
- **Treated: excellent, durable remission** with IL-1 blockade — "Patients treated with anakinra responded rapidly" [PMID 19494218]; ~88% **complete remission** on continued anakinra with mean **Hb +3.18 g/dL**, **ESR –53.4 mm/h**, **CRP –135.45 mg/L** [PMID 38398338].
- **Life expectancy:** near-normal with early, sustained treatment; historically poor if undiagnosed.
- **Complications to prevent (tertiary prevention):** bone deformity, pathologic fracture, growth impairment, respiratory failure, thrombosis; chronic autoinflammation carries a theoretical **AA amyloidosis** risk [Review,

P 32882069

- **Prognostic factors: time-to-diagnosis / time-to-anakinra** (earlier = better), severity/extent of organ involvement, treatment adherence, and residual protein function (hypomorphic alleles milder;

P 32185578

- **Prognostic biomarkers:** CRP/ESR normalization tracks treatment response; no validated molecular predictor beyond genotype.

12. Treatment

Pharmacotherapy — IL-1 blockade (mechanism-based, first-line): - **Anakinra** (recombinant IL-1RA; blocks IL1R1) — **first-line, lifesaving;** rapid response, ~88% remission [PMID 19494218, 38398338]. **NCIT:C1960**. ATC L04AC03. Adverse events: injection-site reactions;

rarely **anaphylaxis**, manageable by **desensitization** [PMID 28503715]. - **Rilonacept** (IL-1 "trap" fusion protein neutralizing IL-1 α / β) — effective alternative [PMID 32819369]. **NCIT:C48411**. - **Canakinumab** (anti-IL-1 β mAb) — successful in a late-onset case at **150 mg SC q6 weeks** [PMID 26100510]; **but** because DIRA involves unopposed **IL-1 α as well as IL-1 β** , selective anti-IL-1 β can be **insufficient or trigger flares** in some patients [PMID 32819369]. **NCIT:C71708**. - **Adalimumab** (anti-TNF) — effective in one late-onset case [PMID 31467740]. Corticosteroids give partial/incomplete control [PMID 32819369].

Pharmacogenomics: none established; therapy is **genotype-rational** (replace/antagonize IL-1) rather than metabolizer-guided.

Advanced therapeutics: No approved gene therapy, cell therapy, or RNA therapy for DIRA; loss-of-function biology makes **gene replacement** a conceptually attractive future approach (not yet clinical).

Surgical/interventional: supportive orthopedic management of fractures/deformity; not curative.

Supportive & rehabilitative: pain control, wound/skin care, physical therapy for musculoskeletal sequelae, nutritional support.

Experimental / trials: no DIRA-specific registered pivotal trials (ultra-rare); management is guideline/expert-based and extrapolated from IL-1-mediated autoinflammation.

Treatment strategy: **empiric anakinra** may be started on strong clinical suspicion before genetic confirmation in a deteriorating neonate [PMID 37575641], then transition/optimize once genotype known; **continuous lifelong** IL-1 blockade with dose titration to CRP/ESR and clinical control.

13. Prevention

- **Primary prevention:** not possible (germline monogenic). **Genetic counseling** for at-risk families; **carrier/cascade testing**; **prenatal testing** or **preimplantation genetic diagnosis** feasible when familial variant is known (incl. CNV assays for deletions) [inferred from

P 22431714

P 28503715

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- **Secondary prevention: early recognition** and prompt IL-1 blockade to abort organ damage; targeted testing of neonates with unexplained pustulosis in **founder populations** (e.g., Puerto Rico) [PMID 22431714, 37575641].
- **Tertiary prevention:** sustained IL-1 blockade to prevent flares, bone deformity, thrombosis, and possible amyloidosis [PMID 32882069].
- **Immunization / public-health / behavioral / environmental measures:** not applicable to this monogenic disease.
- **Counseling:** autosomal-recessive **25% recurrence risk** per pregnancy for carrier couples; asymptomatic-carrier reassurance.

14. Other Species / Natural Disease

- **Taxonomy:** experimental disease in **Mus musculus (NCBI:txid10090)**; no reported spontaneous natural DIRA-equivalent in companion animals or wildlife (OMIA: not established).
- **Orthologous gene:** mouse **Il1rn (NCBI Gene 16181)**; human **IL1RN (3557)** — evolutionarily conserved IL-1 antagonism.
- **Natural disease / veterinary relevance:** no recognized naturally occurring animal DIRA; not zoonotic; no cross-species transmission (genetic disease).
- **Comparative biology:** *Il1rn*-null mice reproduce the core biology (IL-1-driven sterile inflammation of skin, joints, bone, arteries), confirming mechanistic conservation, but disease pattern is **strain/background-dependent** [Model organism,

P 15086551

P 21414240

15. Model Organisms

- **Principal model: IL-1Ra (Il1rn) knockout mouse** [Model organism]. "Interleukin-1 receptor antagonist knockout (IL-1Ra KO) mice spontaneously develop aortitis, arthritis and

dermatitis, and are employed as a model for human inflammatory diseases" [PMID 39600116].

- **Genetic model types:** constitutive **knockout** (BALB/c, DBA/1, C57BL/6 backgrounds); congenic QTL strains; transferable T-cell disease (into nude mice) [PMID 39600116, 21414240].
 - **Phenotype recapitulation (strengths):** IL-1-dependent **arthritis, psoriasiform pustular dermatitis with subcorneal neutrophil microabscesses, arterial inflammation/aortitis, and bone changes** — mirroring human skin+bone+systemic disease [PMID 15086551, 16622029, 39191800, 39600116].
 - **Limitations:** disease penetrance is **background-dependent** (BALB/c susceptible, C57BL/6 resistant), governed by **QTLs (chr 1 major locus; candidates *Mr1*, *Pla2g4a*, *Fasl*, *Prg4*, *Ptgs2*)** and by the **microbiome** — features not central to the human monogenic disease [PMID 21414240, 22820384, 22942082, 28645307]; mice do not fully replicate neonatal multifocal osteomyelitis.
 - **Applications:** dissecting IL-1→NF-κB→neutrophil biology, osteoimmunology, microbiome–autoinflammation interactions, and preclinical IL-1-blockade/target validation (e.g., PLD1 inhibition;
- P 23689131
- **Resources:** MGI, IMSR (IL1rn alleles); Alliance of Genome Resources.

Supported vs Refuted Hypotheses

Supported: - H1: DIRA is caused by biallelic LOF *IL1RN* variants producing non-secreted IL-1RA and IL-1 hyperresponsiveness — **strongly supported** [PMID 19494218]. - H2: Unopposed IL-1 → NF-κB → CXC-chemokine/neutrophil axis drives pustulosis and sterile osteomyelitis — **supported** [PMID 16622029, 29742056, 15086551, 33314777]. - H3: IL-1 blockade (anakinra) achieves rapid, durable remission — **supported** (~88% remission) [PMID 38398338, 19494218]. - H4: Founder alleles explain geographic clustering — **supported** [PMID 22431714, 28503715, 32819369].

Refuted / not supported: - That selective IL-1β blockade is uniformly sufficient — **not supported**; it can be insufficient/flare-inducing because IL-1α is also unopposed [PMID 32819369]. - That an environmental/infectious agent causes DIRA — **refuted**; disease is sterile and monogenic [PMID 21788901].

Limitations and Future Directions

- **Ultra-rarity** → no prevalence/incidence figures, no QoL instruments, no controlled trials; evidence is aggregated case-level.
 - **Human modifiers, epigenetics, and omics** are essentially unstudied; mouse QTL/microbiome findings need human validation.
 - Future: DIRA registries and natural-history studies; prospective comparison of anakinra vs rilonacept vs canakinumab; gene-replacement/editing as a potential curative approach; standardized diagnostic criteria and inclusion of *IL1RN* (with CNV detection) in relevant newborn/at-risk screening.
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Key References (PMIDs)

19494218 (landmark discovery), 40060136 (systematic review), 38398338 (anakinra meta-analysis), 22431714 & 22431772 (Puerto Rican 175-kb deletion), 28503715 (Indian 22.5-kb deletion), 21792839 (compound het E77X + c.140delC), 32819369 (Brazilian variants; IL-1 β -blockade caveat), 26100510 (late-onset R26X, canakinumab), 31467740 (late-onset, adalimumab), 32185578 (hypomorphic reinitiation allele), 16622029 (NF- κ B mechanism), 29742056 & 27890235 (IL-1 β /neutrophil dermatologic mechanism), 15086551 (psoriasiform KO skin), 39600116 (KO multi-organ inflammation), 39191800 (KO bone), 21414240/22820384/22942082 (QTL modifiers), 28645307 (microbiome), 33314777 (DIRA vs Majeed/NOMID monocyte/osteoclast biology), 21788901 & 30031498 & 39757386 (sterile osteomyelitis/imaging/DDx), 37575641 (early recognition), 32882069 (SAID/amyloidosis).
